

Introduction & Hook

Ask:

“Why does a virtual ball fall down in VR instead of floating in the air?”

Explain:

Because VR systems simulate **real-world physics** like:

- Gravity
- Force
- Motion

This is called **physically based simulation**.

What is Physically Based Simulation?

Basic Concept

Definition:

Physically based simulation is the use of mathematical models to simulate real-world physical behavior in a virtual environment.

Simulated Properties:

- Gravity
 - Velocity
 - Acceleration
 - Force
 - Mass
 - Friction
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Importance in VR:

- Realism
 - Natural interaction
 - Better user immersion
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Physics Engines

What is a Physics Engine?

Definition:

A physics engine is software that simulates physical behavior of objects in real time.

Examples:

- Unity Physics Engine
 - NVIDIA PhysX
 - Bullet Physics
 - Havok Physics
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Functions of Physics Engine:

- Applies gravity
 - Calculates motion
 - Detects collisions
 - Handles collision response
 - Simulates rigid body dynamics
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Types of Physics:

1. **Rigid Body Physics**
 - o Objects do not deform
 - o Example: Ball, box
2. **Soft Body Physics**

- o Objects deform
 - o Example: Cloth, jelly
 - 3. **Particle Systems**
 - o Used for fire, smoke, fluids
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Important Concept:

Newton's Second Law:

$$F=m \times a$$

Force determines acceleration.

Collision Detection

What is Collision Detection?

Definition:

Collision detection determines when two objects intersect or come into contact.

Why Needed:

- Prevent objects from passing through each other
 - Enable interaction
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Types of Collision Detection:

1. Broad Phase

- Fast, approximate

- Uses bounding volumes
- Eliminates unlikely collisions

Example:

- Bounding box overlap
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2. Narrow Phase

- Accurate, detailed
- Uses actual geometry

Example:

- Triangle-level checking
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Example:

VR ball hitting wall:

1. Check bounding box → possible collision
 2. Check exact mesh → confirm collision
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Collision Response

What Happens After Collision?

Definition:

Collision response determines how objects behave after collision.

Types of Responses:

1. Elastic Collision

- Objects bounce
- Energy conserved

Example:

- Bouncing ball
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2. Inelastic Collision

- Objects stick or slow down

Example:

- Clay hitting surface
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3. Friction-Based Response

- Sliding or stopping

Example:

- Box sliding on floor
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Real VR Example

VR game:

- Player throws object
- Physics engine applies gravity
- Object hits floor
- Collision detected
- Object bounces realistically

This creates immersion.

Recap & Questions

Key Points:

- Physics simulation makes VR realistic
 - Physics engines handle motion and forces
 - Collision detection identifies object contact
 - Collision response defines behavior after impact
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Quick Questions:

1. What is a physics engine?
 2. What is collision detection?
 3. Difference between broad-phase and narrow-phase?
 4. What is elastic collision?
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1 Numerical: Newton's Second Law

Formula

$$F = m \times a$$

Problem

A virtual object of mass **2 kg** is pushed with a force of **10 N**.
Find the acceleration.

Solution

$$a = \frac{F}{m} = \frac{10}{2} = 5 \text{ m/s}^2$$

✓ Acceleration = 5 m/s²

Teaching Point

Physics engines use this to compute motion of objects in VR.

2 Numerical: Gravity Effect

Formula

$$F = m \times g$$

Where $g = 9.8 \text{ m/s}^2$

Problem

A VR ball has mass 3 kg.

Find the force due to gravity.

Solution

$$F = 3 \times 9.8 = 29.4 \text{ N}$$

✓ Gravitational force = 29.4 N

Teaching Point

This is why objects fall in VR environments.

3 Numerical: Velocity After Time

Formula

$$v = u + at$$

Problem

A virtual object starts from rest ($u = 0$)

Acceleration = 2 m/s^2

Time = 5 seconds

Find final velocity.

Solution

$$v = 0 + (2 \times 5) = 10 \text{ m/s}$$

✓ Final velocity = 10 m/s

Numerical: Collision Response (Coefficient of Restitution)

Formula

$$e = \frac{v_{after}}{v_{before}}$$

Problem

A ball hits the ground with velocity **10 m/s**

After bouncing, velocity becomes **6 m/s**

Find coefficient of restitution.

Solution

$$e = \frac{6}{10} = 0.6$$

 **e = 0.6**

Interpretation

- $e = 1 \rightarrow$ perfectly elastic
- $e < 1 \rightarrow$ energy loss

5 Numerical: Momentum in Collision

Formula

$$p = m \times v$$

Problem

Object mass = 4 kg

Velocity = 3 m/s

Find momentum.

Solution

$$p = 4 \times 3 = 12 \text{ kg} \cdot \text{m/s}$$

✓ Momentum = 12 kg·m/s

6 Numerical: Broad-Phase Collision Check

Problem

Two objects have bounding spheres:

- Object A radius = 2 units
- Object B radius = 3 units
- Distance between centers = 4 units

Will they collide?

Solution

Collision condition:

$$Distance \leq (R1 + R2)$$

$$4 \leq (2 + 3) = 5$$

✓ Condition satisfied → Collision occurs

Teaching Point

This is how **fast collision detection** works in VR.

7 Numerical: Energy Loss in Collision

Formula

$$KE = \frac{1}{2}mv^2$$

Problem

Mass = 2 kg

Velocity before collision = 10 m/s

Velocity after collision = 5 m/s

Find energy loss.

Solution

Before collision:

$$KE_1 = \frac{1}{2} \times 2 \times (10)^2 = 100J$$

After collision:

$$KE_2 = \frac{1}{2} \times 2 \times (5)^2 = 25J$$

Energy loss:

$$100 - 25 = 75J$$

✓ Energy lost = 75 J